HOW FAST CAN A SET WITH NO INTEGER RATIOS GROW? A SHARP ANSWER TO ERDŐS PROBLEM #1195

CLAUDE OPUS 5.5

ABSTRACT. Let $S \subset (0,\infty)$ be a measurable set such that x/y is never an integer for distinct $x, y \in S$, and let $A_S(x)$ be the Lebesgue measure of $S \cap (0,x)$. Erdős asked how fast $A_S(x)$ can tend to infinity, both for all large x and along a sequence. We show that for a non-decreasing $F \to \infty$ there is such a set with $A_S(x) \geq F(x)$ for all large x if and only if $\int_1^\infty F(x)\, x^{-2}\, dx < \infty$. For an arbitrary $F\colon [1,\infty) \to [0,\infty)$, domination along some sequence $x_m \to \infty$ is possible if and only if $\liminf_{x\to\infty} F(x)/x = 0$. Both statements hold with the integers replaced by any locally finite set of forbidden ratios that contains $\{r^m : m \geq 1\}$ for some $r > 1$, and the constructions work for every locally finite set of forbidden ratios. The upper bound is a fundamental-domain argument for the measure dt/t. The construction, in the streamlined form found by T. Tao, uses Bohr sets in logarithmic coordinates whose frequencies come from Dirichlet's simultaneous approximation theorem.

About this version. This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from `https://boonsuan.github.io/erdos1195_old.pdf`.** The theorems are those of the original. The main construction is now presented in the streamlined Bohr-set form that T. Tao posted on the erdosproblems.com forum, with all details supplied; the classical tools are named and cited (Dirichlet's theorem, Bohr sets, Fejér's lemma), and figures and historical background have been added.

1. INTRODUCTION

1.1. **The problem.** In his 1980 survey of combinatorial number theory, Erdős recorded the following question of W. M. Schmidt [14], [8, p. 101]:

> *Is there a set S of real numbers of infinite measure such that x/y is never an integer for distinct $x, y \in S$?*

Haight [10] and Szemerédi [16] independently constructed such sets, and Haight [11] later showed that one can avoid any prescribed countable set of ratios having no finite limit point. Only the positive part of S matters below, and a subset of an admissible set is admissible. So throughout, $S \subset (0,\infty)$ is measurable, $|\cdot|$ denotes Lebesgue measure, and

$$A_S(x) := |S \cap (0,x)|$$

(Erdős writes $m(S,x)$). Erdős remarked that it is "easy to see" that $A_S(x) = o(x)$, and asked how fast $A_S(x)$ can tend to infinity. More precisely he posed two questions [8, p. 101]: given $F(x) \to \infty$, is there such a set with $A_S(x) > F(x)$ for all $x > x_0$, or only along a sequence $x_m \to \infty$? He predicted that the two answers would be quite different. This is Problem #1195 on Bloom's website [3].

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1.2. **The answer.** Both of Erdős's questions have clean answers, and they are indeed quite different.

Theorem 1.1 (eventual growth)**.** *Let $F\colon [1,\infty)\to[0,\infty)$ be non-decreasing with $F(x)\to\infty$. There is a measurable set $S\subset(0,\infty)$ of infinite measure such that $x/y\notin\mathbb{Z}$ for all distinct $x,y\in S$ and $A_S(x)\geq F(x)$ for all sufficiently large x, if and only if*

$$\int_1^\infty \frac{F(x)}{x^2}\,dx<\infty. \tag{1}$$

For example, $A_S(x)\geq x/(\log x)^{1+\varepsilon}$ (for large x) is attainable for every $\varepsilon>0$, as is $x/(\log x\,(\log\log x)^{1+\varepsilon})$, but $A_S(x)\geq x/\log x$ is not. Erdős asked for the strict inequality $A_S(x)>F(x)$; since (1) holds for F if and only if it holds for $F+1$, this makes no difference.

Theorem 1.2 (growth along a sequence)**.** *Let $F\colon[1,\infty)\to[0,\infty)$ be any function. There are a measurable set $S\subset(0,\infty)$ of infinite measure with $x/y\notin\mathbb{Z}$ for distinct $x,y\in S$ and a sequence $x_m\to\infty$ with $A_S(x_m)\geq F(x_m)$ for all m, if and only if* $\liminf_{x\to\infty}F(x)/x=0$.

So along a sequence any $F=o(x)$ can be beaten (indeed any F that is $o(x)$ merely along some sequence), whereas for all large x the threshold is the integral test (1), which separates $x/\log x$ from $x/(\log x)^{1+\varepsilon}$. The necessity half of Theorem 1.2 is Erdős's observation $A_S(x)=o(x)$.

1.3. **Other forbidden ratios.** The construction extends to every locally finite set of forbidden ratios; the upper bound extends when the forbidden ratios contain all positive powers of one number. For $\Gamma\subset(1,\infty)$ call a set $S\subset(0,\infty)$ Γ-*free* if $x/y\notin\Gamma$ for all $x,y\in S$. (Forbidding a ratio $g<1$ between two points is the same as forbidding $1/g>1$, so there is no loss in taking $\Gamma\subset(1,\infty)$; and since $1\notin\Gamma$ we may allow $x=y$.) We call Γ *locally finite* if $\Gamma\cap[1,T]$ is finite for every finite $T>1$; in particular, accumulation at 1 is excluded. A nonempty locally finite Γ therefore has a least element, which we denote $\lambda_\Gamma>1$. The integer case is $\Gamma=\{2,3,4,\dots\}$, $\lambda_\Gamma=2$.

Theorem 1.3. *Let $\Gamma\subset(1,\infty)$ be nonempty and locally finite.*

(a) *If $F\colon[1,\infty)\to[0,\infty)$ is non-decreasing and satisfies (1), there is a Γ-free set S of infinite measure with $A_S(x)\geq F(x)$ for all sufficiently large x.*
(b) *If $F\colon[1,\infty)\to[0,\infty)$ satisfies $\liminf_{x\to\infty}F(x)/x=0$, there are a Γ-free set S of infinite measure and a sequence $x_m\to\infty$ with $A_S(x_m)\geq F(x_m)$ for all m.*
(c) *If $\Gamma\supset\{r^m:m\geq1\}$ for some $r>1$, the converses hold: every Γ-free S has $A_S(x)=o(x)$, so the conclusion of* (b) *forces $\liminf F(x)/x=0$; and if $A_S(x)\geq F(x)$ for all large x, then (1) holds.*

Theorems 1.1 and 1.2 are the case $\Gamma=\{2,3,4,\dots\}\supset\{2^m:m\geq1\}$. Part (a) is a quantitative form of Haight's theorem [11]. The square ratios $\{n^2:n\geq2\}$, which contain $\{4^m:m\geq1\}$, are another example covered by (c). Some hypothesis like the one in (c) is needed for the upper bounds: for $\Gamma=\{2\}$ the set $\bigcup_{k\geq0}[4^k,2\cdot4^k)$ is Γ-free but $A_S(x)/x$ has liminf $1/3$ and limsup $2/3$ (Remark 7.1, Fig. 3). Nor does it suffice for Γ to contain an infinite geometric progression $\{ar^m\}$; the powers of a single number are needed (Remark 7.2).

1.4. **Sources.** The growth criteria and their first proofs are due to Ho [12], whose results were obtained with GPT-5.4 Pro. The upper bound (Section 3) is a fundamental-domain argument; it proves Erdős's observation $A_S(x)=o(x)$ and also yields the integral obstruction. For the lower bound, the substance of the result, we present the construction in the additive Bohr-set formulation proposed by Tao [17], supplying its details and deriving both growth statements from it. Remark 5.3 compares this with the original construction, and the final section records what is due to whom.

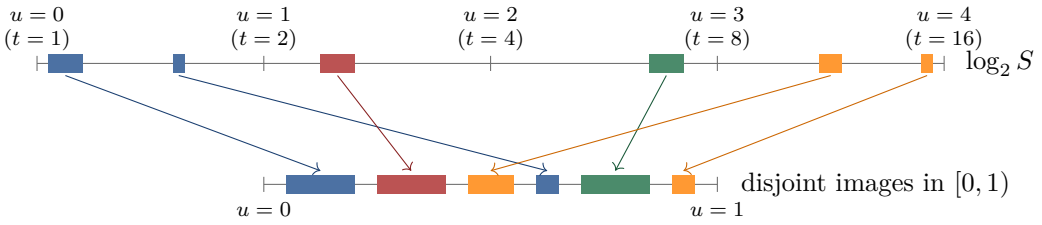


Figure 1. The folding argument of Proposition 3.1, in the coordinate $u = \log_2 t$; equal horizontal steps on the upper axis are doublings of t. If no two points of S have ratio a power of 2, then the pieces $\log_2 S \cap [k, k+1)$ (colours), translated back by k, are pairwise disjoint subsets of $[0, 1)$. Since $dt/t = (\log 2)\, du$, this gives $\int_S dt/t \le \log 2$.

2. The two ideas

Everything becomes additive after taking logarithms. With $u = \log_b t$ for a base $b > 1$, multiplication by γ becomes translation by $\log_b \gamma$, and the measure dt/t becomes $(\log b)\, du$. A Γ-free set of positive reals is then a set $E \subset \mathbb{R}$ with no difference in $\Lambda = \{\log_b \gamma : \gamma \in \Gamma\}$, and A_S is controlled by how much of E lies in each unit interval $[j, j+1)$, weighted by b^j.

Upper bound: folding. If $\Gamma \supset \{2^m : m \ge 1\}$, take $b = 2$. The forbidden differences then include all positive integers, so the pieces $E \cap [k, k+1)$, translated back to $[0, 1)$, must be pairwise disjoint (Fig. 1). Hence E has total length at most 1, that is $\int_S dt/t \le \log 2$. By Tonelli's theorem this integral equals $\int_0^\infty A_S(x) x^{-2}\, dx$, and the integral condition (1) follows.

Lower bound: Bohr sets that almost ignore the forbidden shifts. Now take $b = \lambda_\Gamma$, so that $\Lambda \subset [1, \infty)$ and no two points of the same unit interval $[j, j+1)$ differ by an element of Λ. We build E block by block, using only the blocks $[j, j+1)$ with $j \ge 1$, so that $E \subset [1, \infty)$. The danger is a point y of a later block $[j', j'+1)$ and a point x of an earlier block $[j, j+1)$ with $y - x \in \Lambda$. Since $j \ge 1$, such a difference is less than j', so only the finitely many $\lambda \in \Lambda \cap [1, j']$ can occur. In block j we place $D_j = [j, j+1) \cap \mathrm{B}(\xi_j, \rho_j)$, where

$$\mathrm{B}(\xi, \rho) = \{u \in \mathbb{R} : \|\xi u\| \le \rho\}$$

is a *Bohr set*: a union of short intervals of length $2\rho/\xi$ spaced $1/\xi$ apart, of density 2ρ. Here $\|\cdot\|$ is the distance to the nearest integer. By Dirichlet's theorem we may take the integer frequency $\xi_{j'}$ of a later block so large and so well chosen that $\|\xi_{j'} \lambda\| \le \rho_{j'}$ for every relevant λ: each forbidden shift is almost a period of $\mathrm{B}(\xi_{j'}, \rho_{j'})$. If $y \in D_{j'}$ and $y - x \in \Lambda$, the triangle inequality then forces $x \in \mathrm{B}(\xi_{j'}, 2\rho_{j'})$. So this one thickened Bohr set covers every possible conflict with block j', whichever forbidden shift causes it, and the cost of deleting it from the earlier blocks does not grow with the number of forbidden shifts.

The cost has to be measured relative to the earlier sets D_i, not to the intervals $[i, i+1)$ containing them: D_i is itself thin, and a small fraction of $[i, i+1)$ need not be a small fraction of D_i. But D_i is a fixed finite union of intervals, fixed before $\xi_{j'}$ is chosen, and a Bohr set of very high frequency meets it in about the proportion given by its density. So we can choose $\xi_{j'}$ so large that $\mathrm{B}(\xi_{j'}, 2\rho_{j'})$, of density $4\rho_{j'}$, removes at most $6\rho_{j'}|D_i|$ from each earlier D_i. Summing over all later blocks removes at most $6\sum_{j'} \rho_{j'} \le 3/4$ of D_i if $\sum_j \rho_j \le 1/8$, so at least a quarter of D_i, of measure $\rho_i/2$, survives. Figure 2 shows the first two blocks of this construction for integer ratios, with actual parameters.

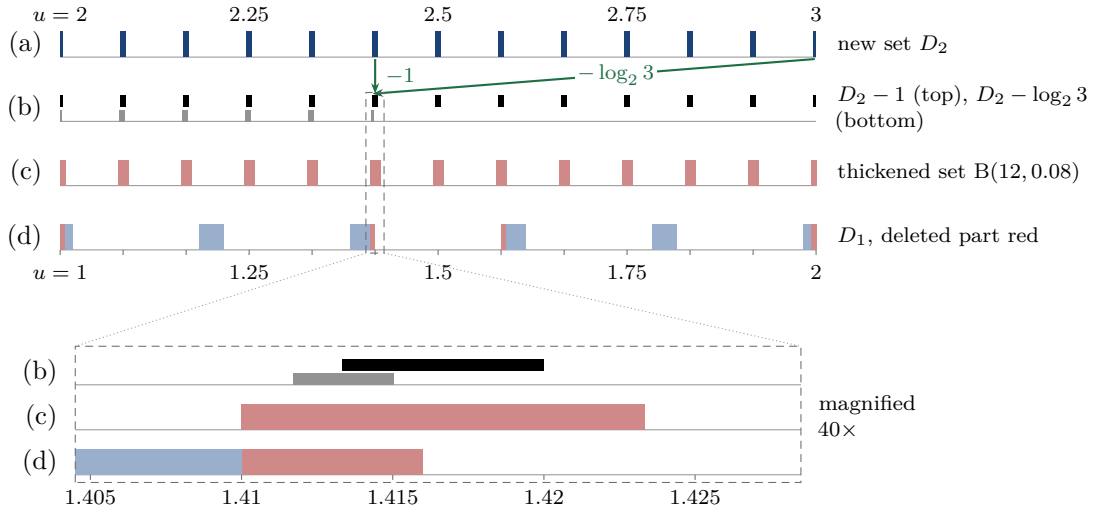


Figure 2. The first two blocks of the construction of Proposition 5.1 for integer ratios, in the coordinate $u = \log_2 t$, with the exact parameters $\xi_1 = 5$, $\rho_1 = 0.08$, $\xi_2 = 12$, $\rho_2 = 0.04$ (so $\rho_1 + \rho_2 \le 1/8$). Row (a) shows the block $[2, 3)$ ($t \in [4, 8)$), rows (b)–(d) the block $[1, 2)$ ($t \in [2, 4)$), each at full width. (a) The new set $D_2 = \mathrm{B}(12, 0.04) \cap [2, 3)$. (b) Its translates by -1 and $-\log_2 3$, in separate sub-rows: the points $x \in [1, 2)$ with $x + 1$ or $x + \log_2 3$ in D_2, the only integer ratios that can link the two blocks. (c) The thickened Bohr set $\mathrm{B}(12, 0.08)$. It contains row (b) because $\|12 \cdot 1\| = 0$ and $\|12 \log_2 3\| \approx 0.0196 \le \rho_2$. (d) The set $D_1 = \mathrm{B}(5, 0.08) \cap [1, 2)$, of length 0.16, after deletion by block 2 only; the deleted part (red) is its intersection with (c). The inset magnifies the dashed window: the arrows follow the two teeth of D_2 whose translates land there (the second is the tooth at $u = 3$, cut off by the end of the block), and both translates lie in one tooth of (c), which overlaps a tooth of D_1. In this example the deleted length is $0.02533\ldots$, close to the limiting-density estimate $4\rho_2 \cdot 0.16 = 0.0256$ and below the required bound $6\rho_2 \cdot 0.16 = 0.0384$.

From block masses to growth rates. The construction lets us prescribe any sufficiently small summable sequence of logarithmic block densities: block j keeps length at least $\rho_j/2$ in the coordinate u, which is ordinary measure at least $(\log b)\rho_j b^j/2$ in $[b^j, b^{j+1})$. For eventual domination we put enough mass into $[b^j, b^{j+1})$ to dominate F throughout the next block, which means taking ρ_j proportional to $F(b^{j+2})/b^j$. For non-decreasing F the integral condition (1) is equivalent to the summability of these weights, and starting sufficiently far out makes their total as small as required. For domination along a sequence we use only selected blocks, choosing observation points x_m where $F(x_m)/x_m$ is so small that the corresponding weights are summable. Compared with the folding bound, avoiding all the other integer ratios costs nothing in eventual growth beyond the integral obstruction already forced by the powers of 2.

3. The upper bound

Proposition 3.1 (folding)**.** *Let $r > 1$ and let $S \subset (0, \infty)$ be measurable with no two points whose ratio lies in $\{r^m : m \ge 1\}$. Then*

$$\int_0^\infty \frac{A_S(x)}{x^2}\, dx = \int_S \frac{dt}{t} \le \log r.$$

Proof. For $k \in \mathbb{Z}$ put $P_k := r^{-k}\big(S \cap [r^k, r^{k+1})\big) \subset [1, r)$. These sets are pairwise disjoint: if $u \in P_i \cap P_k$ with $i < k$, then $r^i u$ and $r^k u$ are points of S with ratio r^{k-i}. The measure dt/t is

invariant under dilations, so substituting $t = r^k u$ on each piece gives

$$\int_S \frac{dt}{t} = \sum_{k\in\mathbb{Z}} \int_{P_k} \frac{du}{u} = \int_{\bigcup_k P_k} \frac{du}{u} \le \int_1^r \frac{du}{u} = \log r.$$

Finally, by Tonelli's theorem,

$$\int_0^\infty \frac{A_S(x)}{x^2}\,dx = \int_0^\infty \int_S \frac{\mathbf{1}_{t<x}}{x^2}\,dt\,dx = \int_S \int_t^\infty \frac{dx}{x^2}\,dt = \int_S \frac{dt}{t}. \qquad \square$$

In the language of harmonic analysis, dt/t is Haar measure on the multiplicative group $(0,\infty)$, the quotient by the subgroup $r^{\mathbb{Z}}$ has total mass $\log r$, and the hypothesis says that S maps injectively to this quotient. The integer analogue is the theory of primitive sequences, in which no term divides another; Erdős discusses these, and a version for real numbers, in the same survey [8, §4]. Besicovitch [2] found primitive sequences of positive upper density, while Behrend [1] and Erdős [7] showed that the lower density is always 0, Erdős by proving that $\sum_{a\in A} 1/(a\log a)$ is bounded by an absolute constant for primitive A. The analogue of $\int_S dt/t$, namely $\sum_{a\in A} 1/a$, can diverge for a primitive set (take the primes). The continuous problem is more rigid.

Corollary 3.2. *Let r and S be as in Proposition 3.1. Then $A_S(x) = o(x)$. If moreover $F\colon [1,\infty) \to [0,\infty)$ is non-decreasing and $A_S(x) \ge F(x)$ for all $x \ge x_0$, then*

$$\int_1^\infty \frac{F(x)}{x^2}\,dx < \infty.$$

Proof. For $0 < R < x$ we have $A_S(x) \le R + |S \cap (R,x)| \le R + x\int_{S\cap(R,x)} dt/t$, since $x/t \ge 1$ there. Hence $\limsup_{x\to\infty} A_S(x)/x \le \int_{S\cap(R,\infty)} dt/t$, which tends to 0 as $R \to \infty$ because $\int_S dt/t < \infty$. For the second statement we may assume $x_0 \ge 1$. Then $\int_{x_0}^\infty F(x)x^{-2}dx \le \log r$ by Proposition 3.1, and $\int_1^{x_0} F(x)x^{-2}dx \le F(x_0) < \infty$. $\square$

4. Two classical tools

Write $\|u\|$ for the distance from $u \in \mathbb{R}$ to the nearest integer. It satisfies $\|-u\| = \|u\|$ and the triangle inequality $\|u+v\| \le \|u\| + \|v\|$. For a frequency $\xi > 0$ and a radius $0 \le r \le 1/2$, the *Bohr set*

$$\mathrm{B}(\xi, r) \coloneqq \{u \in \mathbb{R} : \|\xi u\| \le r\}$$

is the union of the closed intervals $[(n-r)/\xi, (n+r)/\xi]$, $n \in \mathbb{Z}$. Bohr sets go back to Bohr's theory of almost periodic functions [4] and are now a basic tool of additive combinatorics [18]. Here we only need Bohr sets with one frequency, in the group $\mathbb{R}$.

The first tool says that finitely many given numbers can be made simultaneously almost periods of a Bohr set with large integer frequency.

Lemma 4.1 (Dirichlet's simultaneous approximation theorem)**.** *Let $\theta_1, \dots, \theta_m \in \mathbb{R}$ and let $N, Q \ge 1$ be integers. There is an integer ξ with $Q \le \xi \le QN^m$ and $\|\xi\theta_i\| < 1/N$ for $i = 1, \dots, m$.*

Proof. Dirichlet's theorem [6] (see also [5, Ch. I] or [15, Ch. II, §1]) says that for any reals $\varphi_1, \dots, \varphi_m$ there is an integer $1 \le q \le N^m$ with $\|q\varphi_i\| < 1/N$ for all i. For completeness, here is the pigeonhole proof. Divide $[0,1)^m$ into N^m boxes $\prod_i [a_i/N, (a_i+1)/N)$. Two of the N^m+1 points $(\{q\varphi_1\}, \dots, \{q\varphi_m\})$, $0 \le q \le N^m$, lie in the same box, say for $q_1 < q_2$, and $q = q_2 - q_1$ works. Apply this with $\varphi_i = Q\theta_i$ and put $\xi = qQ$. $\square$

The second tool says that a Bohr set of very high frequency meets every fixed finite union of intervals in the proportion given by its density.

Lemma 4.2 (equidistribution of high-frequency Bohr sets)**.** *Let $A \subset \mathbb{R}$ be a finite union of bounded intervals and let $0 \le r \le 1/2$. Then*

$$|A \cap \mathrm{B}(\xi, r)| \longrightarrow 2r|A| \qquad (\xi \to \infty).$$

Proof. Write A as a disjoint union of n intervals. It suffices to show that a single interval I satisfies $\big||I \cap \mathrm{B}(\xi, r)| - 2r|I|\big| < 1/\xi$. The set $\mathrm{B}(1, r)$ is 1-periodic and meets each interval of length 1 in measure $2r$. Let $\ell = \lfloor \ell \rfloor + \vartheta$ be the length of an interval. Partition it, starting at its left endpoint, into $\lfloor \ell \rfloor$ intervals of length 1 and a final interval of length $\vartheta < 1$. By periodicity, each length-1 interval meets $\mathrm{B}(1, r)$ in measure $2r$, so the whole interval meets $\mathrm{B}(1, r)$ in measure $2r\lfloor \ell \rfloor + \eta$ with $0 \le \eta \le \vartheta$. This differs from $2r\ell$ by at most $\vartheta < 1$. Since $\mathrm{B}(\xi, r) = \xi^{-1}\mathrm{B}(1, r)$, scaling by ξ^{-1} gives the claim. □

Lemma 4.2 is the simplest case of a Riemann–Lebesgue-type averaging statement usually called Fejér's lemma: if $f \in L^1(\mathbb{R})$ and g is bounded, measurable and 1-periodic, then $\int f(u)g(\xi u)\,du \to \left(\int_0^1 g\right) \int f$ as $\xi \to \infty$. The statement in this generality is a special case of Corollary 1 of Kahane [13, p. 109], who attributes the case $g(u) = |\sin u|$ to Fejér [9]. We need only finite unions of intervals.

5. The Bohr-set construction

The following proposition, in additive coordinates, carries the whole lower bound. It and the idea of its proof are due to Tao [17], who stated it for a discrete Λ and positive weights; we allow zero weights, which is convenient for Theorem 1.3(b).

Proposition 5.1 (Tao)**.** *Let $\Lambda \subset [1, \infty)$ be locally finite, and let $\rho_1, \rho_2, \dots \ge 0$ satisfy $\sum_{j \ge 1} \rho_j \le 1/8$. Then there is a Borel set $E \subset [1, \infty)$ such that $(E - E) \cap \Lambda = \varnothing$ and*

$$|E \cap [j, j+1)| \ge \rho_j/2 \qquad \text{for every } j \ge 1.$$

Proof. Let $J = \{j \ge 1 : \rho_j > 0\}$. For $j \in J$ we shall choose integer frequencies $\xi_j \ge 1$ and put

$$D_j := [j, j+1) \cap \mathrm{B}(\xi_j, \rho_j), \qquad E_j := D_j \setminus \bigcup_{j' \in J,\ j' > j} \mathrm{B}(\xi_{j'}, 2\rho_{j'}), \qquad E := \bigcup_{j \in J} E_j.$$

So D_j is the Bohr set placed in block j, and E_j is what is left of it after every later block has removed its thickened Bohr set. Since ξ_j is an integer, $x \mapsto \xi_j x$ maps $[j, j+1)$ onto ξ_j full periods of the 1-periodic set $\mathrm{B}(1, \rho_j)$, and $\rho_j \le 1/8 < 1/2$, so D_j is a finite union of intervals with

$$|D_j| = 2\rho_j. \tag{2}$$

Choice of frequencies. Go through J in increasing order. When ξ_i is already defined for all $i \in J$ with $i < j$, choose the integer ξ_j so that

(F1) $\|\xi_j \lambda\| \le \rho_j$ for every $\lambda \in \Lambda \cap [1, j]$;
(F2) $|D_i \cap \mathrm{B}(\xi_j, 2\rho_j)| \le 6\rho_j |D_i|$ for every $i \in J$ with $i < j$.

Condition (F1) contains the conflicts; condition (F2) bounds the loss of measure. Both can be met. The earlier sets D_i are fixed finite unions of intervals of positive measure, and $2\rho_j \le 1/4$, so by Lemma 4.2 the left side of (F2) tends to $4\rho_j|D_i|$ as $\xi_j \to \infty$; the constant 6 leaves room for the error at finite ξ_j. Hence (F2) holds, for these finitely many i, for all integers ξ_j exceeding some Q. The set $\Lambda \cap [1, j]$ is finite, so Lemma 4.1, with $N \ge 1/\rho_j$ and this Q, provides an integer $\xi_j \ge Q$ that also satisfies (F1).

Measurability. Each D_j is a finite union of intervals and each $\mathrm{B}(\xi, r)$ is closed, so every E_j, and hence E, is Borel.

Size. Fix $j \in J$. By (F2) applied to each later block j' (with $i = j$), and since $\sum_{j'>j} \rho_{j'} \le 1/8$,

$$|E_j| \ge |D_j| - \sum_{j' \in J,\, j'>j} |D_j \cap \mathrm{B}(\xi_{j'}, 2\rho_{j'})| \ge |D_j|\Big(1 - 6\sum_{j'>j} \rho_{j'}\Big) \ge \tfrac{1}{4}|D_j| = \tfrac{\rho_j}{2}.$$

As $E \cap [j, j+1) = E_j$ for $j \in J$, this is the required bound; for $j \notin J$ there is nothing to prove.

No forbidden differences. Suppose that $x, y \in E$ and $\lambda := y - x \in \Lambda$. Let $x \in E_j$ and $y \in E_{j'}$. Two points of the same block differ by less than $1 \le \lambda$, so $j \ne j'$, and since $y > x$ we have $j < j'$. Moreover

$$1 \le \lambda = y - x < (j' + 1) - j \le j', \tag{3}$$

where the last step uses $j \ge 1$. Thus $\lambda \in \Lambda \cap [1, j']$, and (F1) gives $\|\xi_{j'}\lambda\| \le \rho_{j'}$. Also $\|\xi_{j'} y\| \le \rho_{j'}$, because $y \in D_{j'}$. By the triangle inequality,

$$\|\xi_{j'} x\| = \|\xi_{j'} y - \xi_{j'}\lambda\| \le \|\xi_{j'} y\| + \|\xi_{j'}\lambda\| \le 2\rho_{j'},$$

so $x \in \mathrm{B}(\xi_{j'}, 2\rho_{j'})$ with $j' \in J$ and $j' > j$. This contradicts $x \in E_j$. Hence $(E - E) \cap \Lambda = \varnothing$. □

Remark 5.2 (two features of the proof). The range calculation (3) uses $j \ge 1$, that is, $E \subset [1, \infty)$; this explains the choice to start with the block $[1, 2)$. (A block $[0, 1)$ would only require (F1) for the still finite set $\Lambda \cap [1, j' + 1)$.) The measure estimate uses only the equidistribution of each later Bohr set against each earlier D_i, one pair at a time, and then a union bound. No joint equidistribution of the frequencies is needed.

Remark 5.3 (comparison with the original argument). The original note [12] ran the same scheme multiplicatively, proving its tools from scratch. It worked with a base $1 < b < \lambda_\Gamma$ and open annuli (b^j, b^{j+1}), and put in a new annulus the set $H_j = (b^j, b^{j+1}) \cap \{t > 0 : \|q \log_b t\| < \alpha\}$, the restriction to that annulus of the exponential image of a Bohr set; it called these sets logarithmic combs. Its frequency q came from a lemma on "large simultaneous returns", proved via compactness of the orbit closure in the torus, which is a qualitative form of Lemma 4.1, applied to the finitely many ratios that can connect H_j to earlier blocks. A "periodic averaging" lemma, a bounded-interval version of Fejér's lemma with a bounded Riemann-integrable periodic factor, played the role of Lemma 4.2. After all blocks were chosen, the note deleted from each earlier block only its Γ-*shadow* $\{x : \gamma x \in H_j \text{ for some } \gamma \in \Gamma\}$, which it showed to lie in a thickened comb. Tao's version deletes the whole thickened Bohr set, so shadows never need to be computed, and takes $b = \lambda_\Gamma$ itself, which works because the blocks are half-open. The quantitative outcomes are the same up to constants.

6. Proofs of the theorems

We first translate Proposition 5.1 back to multiplicative language.

Corollary 6.1. *Let $\Gamma \subset (1, \infty)$ be nonempty and locally finite, put $b := \lambda_\Gamma$, and let $\rho_1, \rho_2, \dots \ge 0$ satisfy $\sum_j \rho_j \le 1/8$. There is a Γ-free Borel set $S \subset [b, \infty)$ with*

$$|S \cap [b^j, b^{j+1})| \ge \tfrac{\log b}{2}\, \rho_j\, b^j \qquad (j \ge 1).$$

Proof. Let $\Lambda := \{\log_b \gamma : \gamma \in \Gamma\}$. Since $b = \min \Gamma$ we have $\Lambda \subset [1, \infty)$, and $\Lambda \cap [1, T] = \log_b(\Gamma \cap [b, b^T])$ is finite, so Λ is locally finite. Let E be the set given by Proposition 5.1 and put $S := \{b^u : u \in E\}$. This is a Borel subset of $[b, \infty)$, because $u \mapsto b^u$ is a homeomorphism.

For $s = b^u$ and $t = b^v$ in S we have $s/t = b^{u-v}$, which lies in Γ if and only if $u - v \in \Lambda$. So S is Γ-free. Finally,

$$|S \cap [b^j, b^{j+1})| = \int_{E \cap [j,j+1)} b^u \log b \, du \geq b^j \log b \, |E \cap [j, j+1)| \geq \tfrac{\log b}{2} \rho_j b^j. \qquad \square$$

Proof of Theorem 1.3(a). The construction gives each block enough mass to beat F throughout the *next* block; the price, a factor b^2 in the weights, is harmless. Replacing $F(x)$ by $F(x) + \log x$ preserves monotonicity and (1), so we may assume $F \to \infty$. Let $b = \lambda_\Gamma$. Since F is non-decreasing,

$$\int_{b^\ell}^{b^{\ell+1}} \frac{F(x)}{x^2}\, dx \geq F(b^\ell)\Big(\frac{1}{b^\ell} - \frac{1}{b^{\ell+1}}\Big) = \Big(1 - \frac{1}{b}\Big)\frac{F(b^\ell)}{b^\ell},$$

so $\sum_{\ell \geq 0} F(b^\ell)/b^\ell < \infty$. Choose $J \geq 1$ with

$$\frac{2b^2}{\log b} \sum_{\ell \geq J+2} \frac{F(b^\ell)}{b^\ell} \leq \frac{1}{8},$$

and set $\rho_j := 0$ for $j < J$ and $\rho_j := 2F(b^{j+2})/(b^j \log b)$ for $j \geq J$. Then

$$\sum_{j \geq 1} \rho_j = \frac{2}{\log b} \sum_{j \geq J} \frac{F(b^{j+2})}{b^j} = \frac{2b^2}{\log b} \sum_{\ell \geq J+2} \frac{F(b^\ell)}{b^\ell} \leq \frac{1}{8},$$

and Corollary 6.1 yields a Γ-free S with

$$|S \cap [b^j, b^{j+1})| \geq F(b^{j+2}) \qquad (j \geq J).$$

Since $F \to \infty$, S has infinite measure. If $x \geq b^{J+1}$, let $N = \lfloor \log_b x \rfloor \geq J + 1$. Then $[b^{N-1}, b^N) \subset (0, x)$, and

$$A_S(x) \geq |S \cap [b^{N-1}, b^N)| \geq F(b^{N+1}) \geq F(x). \qquad \square$$

Together with Corollary 3.2, this shows that avoiding all the other integer ratios costs nothing in eventual growth beyond the integral obstruction already forced by the powers of 2.

Proof of Theorem 1.3(b). We place mass $M_m := \max(F(x_m), m)$ in a complete block below each observation point x_m. The first term gives domination at x_m; the second gives infinite total measure even when F is bounded or vanishes at the chosen points. Let $b = \lambda_\Gamma$ and $\delta := \log b/(16b^2)$. Choose $x_1 < x_2 < \cdots$ inductively, together with the integers $n_m := \lfloor \log_b x_m \rfloor - 1$. Given $x_1, \ldots, x_{m-1}$, take $x_m \geq \max(b^3,\, b^{n_{m-1}+3})$ (with $n_0 = 0$) so large that

$$F(x_m) \leq 2^{-m} \delta x_m \qquad \text{and} \qquad m \leq 2^{-m} \delta x_m.$$

The first condition holds for arbitrarily large x_m because $\liminf F(x)/x = 0$, and the second for all large x_m. Then $n_m \geq 2$, the n_m are strictly increasing, $x_m \to \infty$, and $b^{n_m+1} \leq x_m < b^{n_m+2}$. Set $\rho_{n_m} := 2M_m/(b^{n_m} \log b)$ and let all other ρ_j be 0. The block $[b^{n_m}, b^{n_m+1})$ lies below x_m, and using it costs at most a factor b^2, since $b^{n_m} > x_m/b^2$; the choice of δ then makes the weights sum to at most $1/8$. Indeed, since $M_m \leq 2^{-m}\delta x_m$,

$$\sum_j \rho_j \leq \frac{2b^2}{\log b} \sum_{m \geq 1} \frac{M_m}{x_m} \leq \frac{2b^2 \delta}{\log b} = \frac{1}{8}.$$

Corollary 6.1 gives a Γ-free S with $|S \cap [b^{n_m}, b^{n_m+1})| \geq M_m$. This interval lies in $(0, x_m)$, so $A_S(x_m) \geq M_m \geq F(x_m)$, and $|S| \geq \sum_m M_m \geq \sum_m m = \infty$. $\square$

Proof of Theorem 1.3(c). A Γ-free set has no two points with ratio in $\{r^m : m \geq 1\}$, so Corollary 3.2 applies. It gives $A_S(x) = o(x)$, and also the integral condition whenever $A_S \geq F$ eventually. If $A_S(x_m) \geq F(x_m)$ along $x_m \to \infty$, then $0 \leq F(x_m)/x_m \leq A_S(x_m)/x_m \to 0$, so $\liminf F(x)/x = 0$. □

Proof of Theorems 1.1 and 1.2. A set $S \subset (0, \infty)$ has $x/y \notin \mathbb{Z}$ for distinct $x, y \in S$ exactly when it is Γ-free for $\Gamma = \{2, 3, 4, \dots\}$, since a ratio of distinct positive numbers that is an integer is at least 2. This Γ is locally finite and contains $\{2^m : m \geq 1\}$, so Theorem 1.3 applies. □

7. Further remarks

Remark 7.1 (a single forbidden ratio). Local finiteness of Γ alone does not force $A_S(x) = o(x)$. For $\Gamma = \{2\}$, the set $S = \bigcup_{k \geq 0} [4^k, 2 \cdot 4^k)$ is Γ-free: if $x \in [4^k, 2 \cdot 4^k)$ then $2x \in [2 \cdot 4^k, 4^{k+1})$, which misses S. Here $A_S(4^{k+1}) = \sum_{i \leq k} 4^i = (4^{k+1} - 1)/3$ and $A_S(2 \cdot 4^k) = (4^{k+1} - 1)/3$. On each gap $[2 \cdot 4^k, 4^{k+1}]$ the ratio $A_S(x)/x$ decreases, and on each piece of S it increases. So $\liminf A_S(x)/x = 1/3$ and $\limsup A_S(x)/x = 2/3$ (Fig. 3). In Theorem 1.3(c) it is the powers of r that make the folding argument possible.

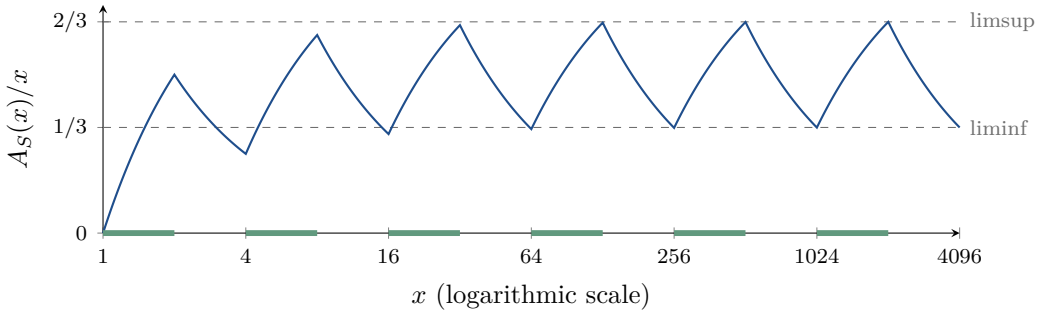


Figure 3. The ratio $A_S(x)/x$ for the $\{2\}$-free set $S = \bigcup_{k \geq 0} [4^k, 2 \cdot 4^k)$, whose pieces are marked in green on the horizontal axis. The ratio rises on each piece and falls on each gap; its liminf is $1/3$ and its limsup $2/3$. When $\Gamma \supset \{r^m : m \geq 1\}$ for some $r > 1$, Corollary 3.2 forces $A_S(x)/x \to 0$ instead.

Remark 7.2 (a geometric progression is not enough). The hypothesis of Theorem 1.3(c) asks for all powers $r, r^2, r^3, \dots$ of a single $r > 1$. Containing an infinite geometric progression does not suffice. Let $\Gamma = \{2 \cdot 4^m : m \geq 0\} = \{2, 8, 32, \dots\}$, a geometric progression containing no set $\{r^m : m \geq 1\}$: every $r \in \Gamma$ is an odd power of 2, and then r^2 is an even power of 2, so $r^2 \notin \Gamma$. The set S of Remark 7.1 is still Γ-free: multiplying a point of $[4^k, 2 \cdot 4^k)$ by $2 \cdot 4^m$ lands in the gap $[2 \cdot 4^{k+m}, 4^{k+m+1})$. So again $\liminf A_S(x)/x = 1/3$; in fact $A_S(x) \geq x/4$ for all $x \geq 4$. Thus $F(x) = x/4$ is dominated for all large x, although $\int_1^\infty F(x) x^{-2}\, dx = \infty$ and $\liminf F(x)/x = 1/4 > 0$; both converses in Theorem 1.3(c) fail.

Remark 7.3 (non-monotone targets). Theorem 1.3(a) assumes F non-decreasing. In general, since A_S is non-decreasing, $A_S(t) \geq F(t)$ for all $t \geq x_0$ implies $A_S(x) \geq F^*(x) := \sup_{x_0 \leq t \leq x} F(t)$ for $x \geq x_0$. So for Γ as in (c), a finite-valued function $F\colon [1, \infty) \to [0, \infty)$, with no monotonicity or measurability assumption, can be dominated eventually if and only if, for some $x_0 \geq 1$, the envelope F^* is finite-valued and $\int_{x_0}^\infty F^*(x) x^{-2}\, dx < \infty$. For sufficiency, apply (a) to the non-decreasing function F^*, extended by the constant $F^*(x_0)$ to $[1, x_0)$.

Acknowledgements and provenance

The results, and the first proofs, are from the note [12] of Boon Suan Ho, obtained with GPT-5.4 Pro prompted by Ho. That note contains Theorems 1.1–1.3, the folding argument of Section 3, the example of Remark 7.1, the treatment of non-monotone targets, and a block construction with logarithmic combs, simultaneous returns and summable deletions (Remark 5.3). The streamlined construction comes from the discussion thread https://www.erdosproblems.com/forum/thread/1195. In his comment of 28 April 2026 [17], Terence Tao recast the argument in logarithmic coordinates and the language of Bohr sets, observed that the note's simultaneous-returns lemma is essentially Dirichlet's simultaneous approximation theorem, and stated Proposition 5.1 with a sketch of its proof: the enlarged Bohr set of each block is deleted from all earlier blocks, and a conditional-density estimate with a union bound controls the loss. The proof of Proposition 5.1 written out here, in which (F2) is an explicit error bound for that estimate and takes the place of an appeal to joint equidistribution, together with the computation (3), the treatment of zero weights, the derivation of Theorem 1.3 from it, and the example of Remark 7.2, are by the present author. The classical ingredients are due to Dirichlet [6], Bohr [4] and Fejér [9], and the history of the problem is as recorded by Erdős [8] and Bloom [3]. This text was written by Claude Opus 5.5.

References

[1] F. Behrend, *On sequences of numbers not divisible one by another*, J. London Math. Soc. **10** (1935), 42–44.
[2] A. S. Besicovitch, *On the density of certain sequences of integers*, Math. Ann. **110** (1934), 336–341.
[3] T. F. Bloom, *Erdős Problem #1195*, https://www.erdosproblems.com/1195, accessed 28 October 2026.
[4] H. Bohr, *Zur Theorie der fastperiodischen Funktionen. I. Eine Verallgemeinerung der Theorie der Fourierreihen*, Acta Math. **45** (1925), 29–127.
[5] J. W. S. Cassels, *An Introduction to Diophantine Approximation*, Cambridge Tracts in Mathematics and Mathematical Physics 45, Cambridge University Press, 1957.
[6] G. Lejeune Dirichlet, *Verallgemeinerung eines Satzes aus der Lehre von den Kettenbrüchen nebst einigen Anwendungen auf die Theorie der Zahlen*, Bericht über die Verhandlungen der Königlich Preußischen Akademie der Wissenschaften zu Berlin (1842), 93–95.
[7] P. Erdős, *Note on sequences of integers no one of which is divisible by any other*, J. London Math. Soc. **10** (1935), 126–128.
[8] P. Erdős, *A survey of problems in combinatorial number theory*, Ann. Discrete Math. **6** (1980), 89–115.
[9] L. Fejér, *Lebesguesche Konstanten und divergente Fourierreihen*, J. Reine Angew. Math. **138** (1910), 22–53.
[10] J. A. Haight, *A linear set of infinite measure with no two points having integral ratio*, Mathematika **17** (1970), 133–138.
[11] J. A. Haight, *A set of infinite measure whose ratio set does not contain a given sequence*, Mathematika **22** (1975), 195–201.
[12] B. S. Ho, *Erdős's ratio-growth problem and a quantitative Haight-type theorem*, note, April 2026, https://boonsuan.github.io/erdos1195_old.pdf.
[13] C. S. Kahane, *Generalizations of the Riemann–Lebesgue and Cantor–Lebesgue lemmas*, Czechoslovak Math. J. **30** (1980), 108–117.
[14] W. M. Schmidt, *Disproof of some conjectures on Diophantine approximations*, Studia Sci. Math. Hungar. **4** (1969), 137–144.
[15] W. M. Schmidt, *Diophantine Approximation*, Lecture Notes in Mathematics 785, Springer, 1980.
[16] E. Szemerédi, *On a problem of W. Schmidt*, Studia Sci. Math. Hungar. **6** (1971), 287–288.
[17] T. Tao, comment of 28 April 2026 on the thread for Erdős Problem #1195, https://www.erdosproblems.com/forum/thread/1195#post-6005.
[18] T. Tao and V. Vu, *Additive Combinatorics*, Cambridge Studies in Advanced Mathematics 105, Cambridge University Press, 2006.

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